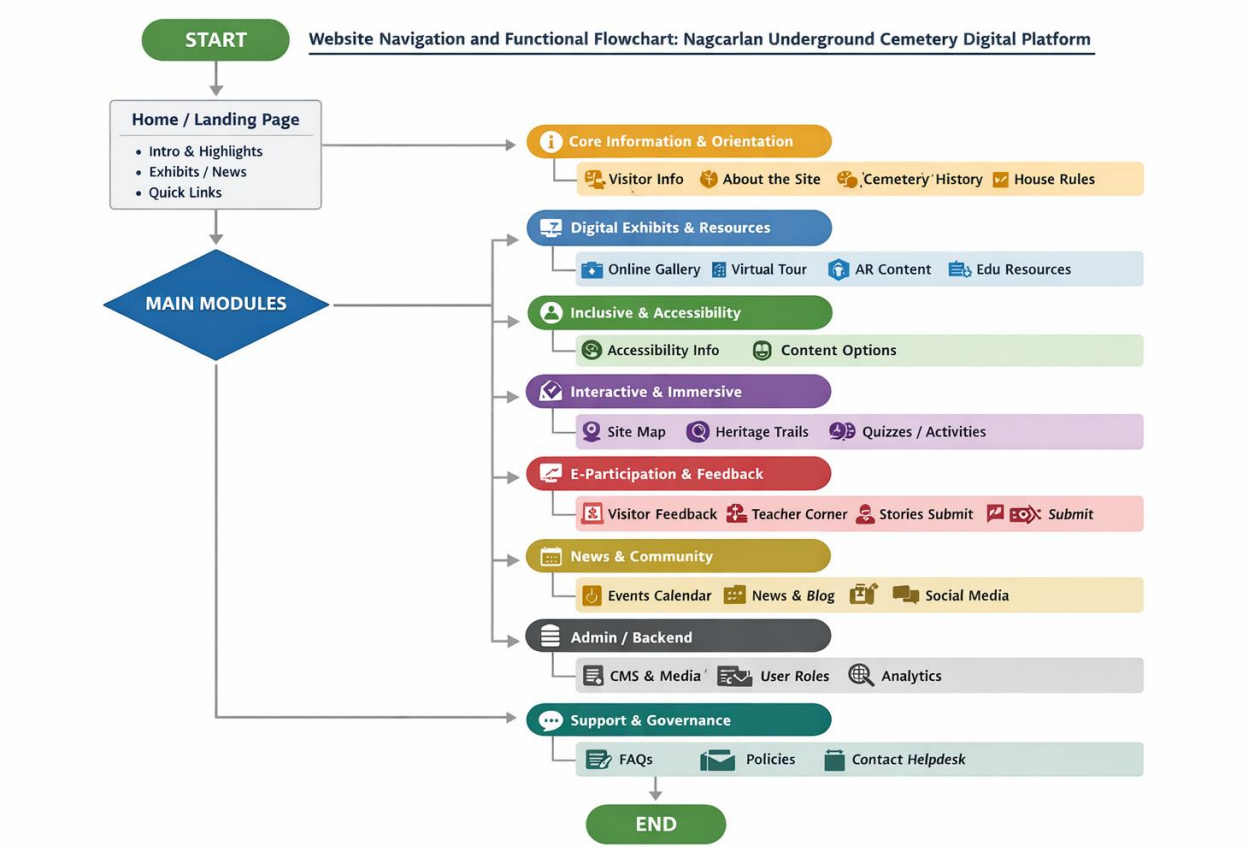
EW01 -Design and Development of a WebXR-Enabled Digital Heritage Experience Platform for the Nagcarlan Underground Cemetery



2. Introduction

The Nagcarlan Underground Cemetery in Laguna is a nationally significant historical and cultural landmark known for its unique underground crypts and its role in Philippine socio-political history. It symbolizes layered narratives of faith, resistance, colonialism, and local community memory. However, access to the site is constrained by geography, cost, mobility barriers, conservation limits, and the lack of richly interactive interpretive materials.

Emerging WebXR technologies allow immersive, browser-based experiences across a wide range of devices, including low- to mid-range smartphones, without installing specialized apps or owning VR headsets. When coupled with inclusive and universal design principles, WebXR can widen participation in cultural heritage by enabling people with disabilities (PWDs),

geographically remote learners, older adults, and low-income communities to experience, understand, and co-create heritage narratives. This study proposes the design and development of an inclusive WebXR-enabled digital heritage platform for the Nagcarlan Underground Cemetery that foregrounds accessibility, educational value, and community participation.

3. Problem Statement

Despite the historical importance of the Nagcarlan Underground Cemetery, current digital representations are limited, fragmented, and primarily text- or image-based. These resources:

- Do not provide an immersive sense of “being there”.
- Are not systematically aligned with curriculum outcomes in history, culture, and ICT.
- Rarely follow accessibility standards, making them difficult to use by PWDs (e.g., low vision, hearing impairments, mobility challenges, neurodivergent users).
- Offer minimal mechanisms for user feedback, data analytics, and community contribution.

Moreover, digital heritage initiatives in the Philippines often overlook inclusive design, local language options, and varying levels of digital literacy and connectivity. There is a gap for a WebXR platform that:

1. Is inclusive and usable across a range of devices and connectivity conditions.
2. Integrates universal design and accessibility principles for diverse users.
3. Embeds culturally grounded narratives co-developed with local communities.
4. Provides evidence of educational and affective impact through systematic evaluation.

4. Objectives

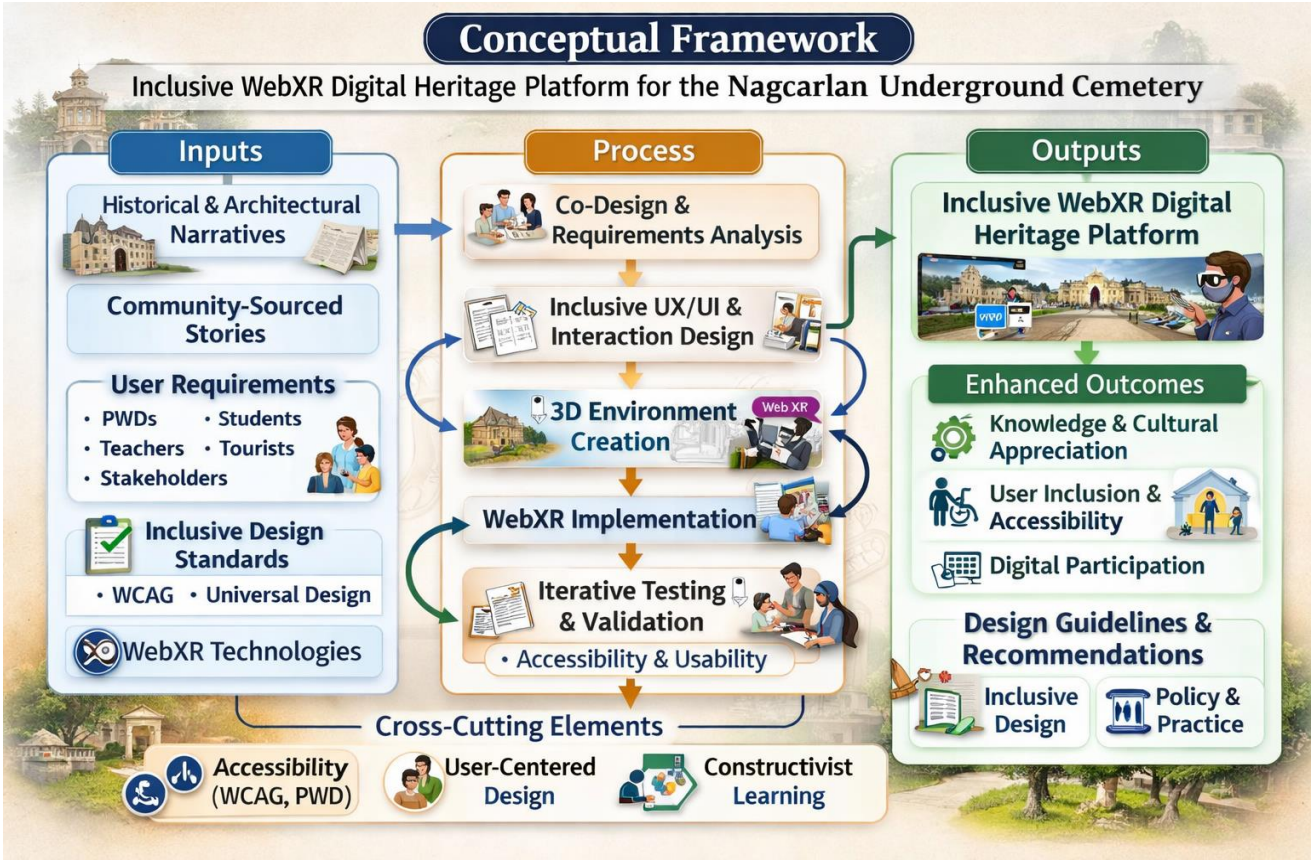
4.1 General Objective

The study aims to develop an integrated, accessible, and interactive web-based platform for the Nagcarlan Underground Cemetery that provides comprehensive information, digital exhibits, and immersive learning experiences; incorporates inclusive and participatory features for diverse audiences, including persons with disabilities; and supports sustainable content management, community engagement, and evidence-based governance in alignment with NHCP policies.

4.2 Specific Objectives

1. To design and develop a modular, accessible web-based platform for the Nagcarlan Underground Cemetery that:
    a. integrates core site information, digital exhibits, virtual tours, AR content, interactive learning activities, and e-participation features for diverse user groups;
    b. incorporates an accessible WebXR-based 3D environment enriched with multimodal content (text, audio, images, sign-language-ready video placeholders, tactile-ready descriptions) and local language support;
    c. embeds narrative-driven, culturally sensitive interactive experiences (guided tours, hotspots, branching stories, reflections, quizzes) that support both formal and informal learning; and
    d. Implement interactive and immersive heritage engagement tools (e.g., digital trails, quizzes, certificates/badges) that foster active user participation, reflection, and sustained interest in the site's history and culture.
2. To implement and assess inclusive interaction, accessibility, and governance mechanisms, including accessible content options, assistive navigation, community storytelling and feedback channels, and an administrative backend for content, users, and analytics.
3. To evaluate the platform's usability, accessibility, user experience, perceived learning, sense of presence, and cultural appreciation among diverse user groups, including PWDs and low-connectivity users.

4. To generate design guidelines and policy/practice recommendations for inclusive WebXR digital heritage initiatives in Philippine and ASEAN contexts.



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5. Research Questions

1. What inclusive design and accessibility requirements should guide the development of a WebXR-based digital heritage platform for the Nagcarlan Underground Cemetery?

2. How can historical, architectural, and community narratives about the site be represented in ways that are culturally grounded, respectful, and accessible to diverse users?

3. How usable, inclusive, and engaging is the developed platform for students, teachers, tourists, PWDs, and community members using different devices and connectivity conditions?

4. To what extent does the platform enhance:

    - users' historical understanding and cultural appreciation;

    - sense of presence and emotional connection to the site;
    - perceived inclusion and digital participation?

5. What design principles and implementation strategies can be derived to guide future inclusive WebXR heritage projects?

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6. Scope and Delimitation

- Focuses on the Nagcarlan Underground Cemetery as a pilot inclusive digital heritage case.
- Targets WebXR deployment via common desktop and mobile browsers; optional support for VR headsets is secondary.
- Emphasizes inclusive and universal design features, with prioritization of critical accessibility functions over high-end photorealism.
- Aims for bilingual content (e.g., English and Filipino) where practical; other local languages will be scoped as recommendations.
- Evaluation will cover a purposive, diverse sample (including PWDs and low-connectivity users) within limited time and resource constraints.

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7. Significance of the Study

- **Heritage agencies and LGUs:** Provides an inclusive, scalable template for digital heritage projects that reach PWDs, remote communities, and younger generations.
- **Educators and learners:** Delivers a curriculum-aligned, immersive platform for history, culture, and ICT, which can be integrated into classroom and online activities.

- **PWDs and marginalized communities:** Expands opportunities for equitable cultural participation and e-participation, addressing physical, economic, and social barriers.
- **Local community and cultural bearers:** Creates space for community voices and stories to be embedded, curated, and updated, strengthening local ownership.
- **Researchers and designers:** Produces empirical evidence and inclusive design guidelines at the intersection of WebXR, eLearning, digital heritage, and accessibility.

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8. Review of Related Literature (expanded outline)

When you write the full RRL, you can structure it as:

1. **Digital Heritage and Inclusive Heritage Interpretation**
    - Virtual museums, 3D reconstructions, and digital storytelling for heritage.
    - Inclusive heritage frameworks, community-based and participatory approaches.
2. **XR/WebXR Applications in Cultural Heritage**
    - Case studies of VR/AR/WebXR in museums, archaeological sites, and historical reconstructions.
    - Comparison of app-based VR versus browser-based WebXR in terms of accessibility and reach.
3. **Inclusive and Universal Design in Digital Systems**
    - Universal Design, Universal Design for Learning (UDL), and WCAG.
    - Accessible multimedia, multimodal interfaces, and assistive technology integration in heritage and education platforms.
4. **Learning Theories in Immersive Environments**
    - Constructivism, experiential learning, situated learning, and presence/immersion.

    - Evidence of learning gains, engagement, and empathy from immersive heritage experiences.

5. **Philippine and Regional Digital Heritage Efforts**

    - Existing digital initiatives in Philippine heritage sites; gaps in inclusivity, language, and evaluation.

    - Southeast Asian perspectives on digital heritage, community participation, and decolonial approaches.

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9. Theoretical and Conceptual Framework

9.1 Theoretical Bases

- **Constructivist and experiential learning theories:** Users construct meaning by exploring the 3D environment, engaging with stories, and reflecting on historical events.

- **Universal Design and Universal Design for Learning:** Guide the creation of flexible, multimodal pathways that accommodate diverse abilities, preferences, and contexts.

- **Presence and immersion theories:** Explain how a sense of being in the virtual Nagcarlan Underground Cemetery can influence engagement, affect, and learning.

- **Technology Acceptance and Digital Inclusion theories:** Frame how perceived usefulness, ease of use, accessibility, and social influence affect users' willingness to adopt WebXR heritage platforms.

9.2 Conceptual Framework (simplified)

- **Inputs:**

    - Historical, architectural, and community-sourced narratives.

    - User requirements from PWDs, students, teachers, tourists, and local stakeholders.
    - Inclusive design and WebXR standards.

- **Process:**
    - Co-design and requirements analysis with diverse stakeholders.
    - Inclusive UX/UI and interaction design.
    - 3D environment creation and WebXR implementation.
    - Iterative testing with emphasis on accessibility and inclusivity.

- **Outputs:**
    - Inclusive WebXR digital heritage platform.
    - Improved user knowledge, cultural appreciation, sense of inclusion, and digital participation.
    - Inclusive design guidelines and policy/practice recommendations.

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10. Methodology

10.1 Research Design

A design and development research approach with mixed methods will be used, emphasizing co-design and participatory methods:

- Iterative cycles of analysis, design, implementation, and evaluation.
- Combination of quantitative (surveys, analytics, tests) and qualitative (interviews, focus groups, observation) data.

10.2 Phases / Procedures

1. **Inclusive Requirements Analysis and Co-Design**
    - Stakeholder mapping to include PWDs, elders, youth, teachers, tourism and heritage officers, and local community representatives.
    - Conduct accessible co-design workshops and interviews (e.g., sign-language interpreters, easy-read materials, remote options).
    - Identify barriers (device, connectivity, literacy, disability) and desired features.
    - Define user personas and use cases representing a wide range of users.
2. **Inclusive System and Experience Design**
    - Draft information architecture, interaction flows, and accessibility options.
    - Produce low- and high-fidelity prototypes, with explicit variations for:
        - Low-bandwidth and low-spec-device modes.
        - High-contrast and large-font modes.
        - Audio-described and captioned content.
    - Validate prototypes with representatives from each target user group.
3. **Development and Integration**
    - Model the cemetery's above-ground chapel and underground crypt using optimized 3D assets.
    - Implement WebXR using frameworks that support progressive enhancement.
    - Integrate:
        - Multilingual text and audio guides.

        - Captions and transcripts for all audio and video.
        - Keyboard and switch-accessible navigation paths.
        - Alternative “guided tour” mode for users with limited XR familiarity.
    - Implement analytics (with privacy safeguards) to capture interaction patterns and accessibility usage.

4. **Testing and Inclusive Evaluation**
    - **Technical and accessibility testing:** Across devices, browsers, and assistive technologies (screen readers where possible).
    - **User testing with diverse groups:**
        - Pre- and post-experience questionnaires on knowledge, cultural appreciation, and self-reported inclusion.
        - Standardized usability scales (e.g., SUS) plus accessibility and inclusion scales.
        - Think-aloud sessions and observation to identify pain points.
    - **Qualitative evaluation:** Focus groups and interviews with PWDs, students, teachers, and community members on perceptions of respectfulness, representation, and empowerment.

5. **Refinement, Documentation, and Knowledge Sharing**
    - Prioritize fixes that improve accessibility and inclusivity.
    - Document design patterns, challenges, and workarounds relevant to low-resource, Global South contexts.
    - Prepare training materials (videos, guides) for teachers and LGU staff to adopt and maintain the platform.

10.3 Participants and Sampling

- **Participants:**
    - Students (SHS, undergraduate, possibly graduate) from different disciplines.
    - Teachers/educators (history, social science, ICT, SPED).
    - PWDs (visual, hearing, mobility, and neurodivergent participants, as feasible and ethical).
    - LGU, tourism, and heritage staff; local community representatives.
- **Sampling:**
    - Purposive sampling to ensure diversity across abilities, age, and digital access.
    - Target numbers (to be refined based on ethics and resources), e.g.:
        - 60–100 learners for quantitative analysis.
        - 20–30 PWD participants across disability types (in collaboration with PDAO/DRRMO/SPED units).
        - 10–20 key informants and experts for qualitative data.

10.4 Data Gathering Instruments

- Demographic and background survey with inclusive questions (e.g., disability, access to devices, language preferences).
- Structured usability and accessibility questionnaires (e.g., SUS + custom inclusion items).
- Pre-/post-tests on knowledge and cultural appreciation (with accessible formats).
- Observation checklists emphasizing accessibility and inclusion.
- Semi-structured interview and focus group guides tailored for different user groups and using accessible formats (easy-read, audio prompts, etc.).

11. Data Analysis

- **Quantitative:**
    - Descriptive statistics for usability, accessibility, presence, learning, and inclusion indicators.
    - Paired comparisons for knowledge and appreciation (pre vs post).
    - Group comparisons (e.g., across device types, disability groups) if sample sizes permit.
- **Qualitative:**
    - Thematic analysis focusing on experiences of inclusion/exclusion, representation, usability barriers, and suggestions.
    - Triangulation between user feedback, analytics logs, and observation data to refine inclusive design guidelines.

12. Ethical and Inclusion Considerations

- Partner with disability-focused offices/organizations to design safe, respectful, and accessible engagements.
- Obtain informed consent/assent, with accessible information sheets and multiple modes (text, audio, sign-language interpreted).
- Ensure voluntary participation, confidentiality, and the right to withdraw without consequences.
- Avoid tokenistic representation by involving PWDs and local communities throughout the design and evaluation process, not only as test subjects.

- Present the digital environment as an interpretive and educational representation, acknowledging cultural sensitivities and local beliefs.

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13. Expected Outputs

- An inclusive, browser-based WebXR platform for the Nagcarlan Underground Cemetery, with documented accessibility and inclusivity features.
- Empirical evidence on inclusive WebXR design for heritage, including data on usability, accessibility, learning, and cultural appreciation.
- A set of context-aware inclusive design guidelines for WebXR in low- to middle-income country settings.
- Training and extension materials for LGUs, schools, and community groups to adapt and maintain inclusive digital heritage platforms.
- Academic publications and presentations in digital heritage, HCI, eLearning, and inclusion/assistive technology forums.

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14. Timetable (example, 12–15 months)

You can stretch or compress as needed:

- Months 1–3: Stakeholder engagement, inclusive requirements analysis, co-design workshops.
- Months 4–5: System and experience design; accessibility and language planning.
- Months 6–9: Development, integration of inclusive features, initial testing.
- Months 10–12: Inclusive evaluation, data collection, data analysis.
- Months 13–15: Refinement, documentation, dissemination, and training.